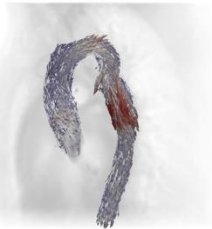


Selection of 4D Flow MRI velocity data

(I) 4D Flow MRI velocity field

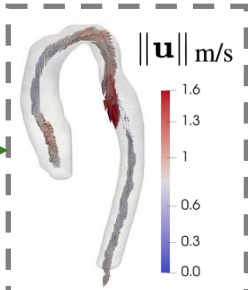


(II) Centerline-based tubular training region

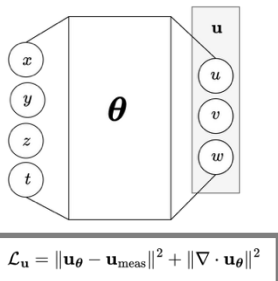


Stage 1: Velocity-field reconstruction and momentum-component estimation

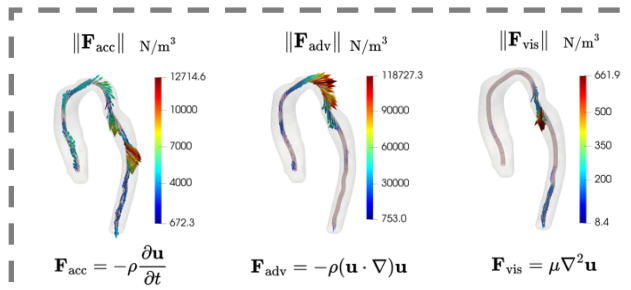
(a) Selected 4D Flow MRI training data



(b) Velocity network

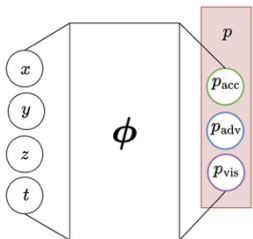


(c) Momentum-component estimation via automatic differentiation



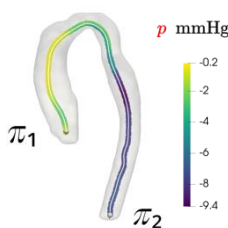
Stage 2: Pressure-component reconstruction and pressure-drop evaluation

(d) Pressure-component network



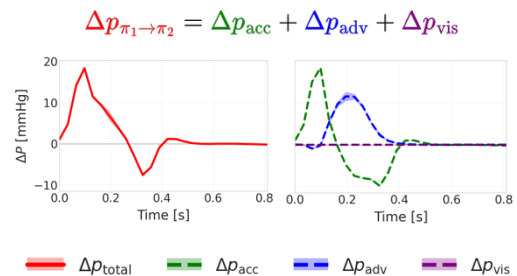
$$\mathcal{L}_p = \sum_{j \in \{\text{acc}, \text{adv}, \text{vis}\}} \|\nabla p_{j, \phi} - \mathbf{F}_j\|^2$$

(e) Reconstructed relative pressure field and evaluation locations



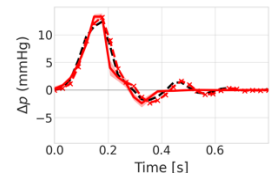
$$p = p_{\text{acc}} + p_{\text{adv}} + p_{\text{vis}}$$

(f) Decomposed pressure-drop waveforms



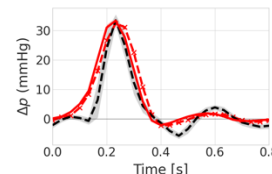
Validation across translational testbeds

In silico



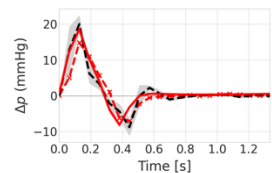
$$\Delta p^{\text{PINN}} \quad \Delta p^{\text{vWERP}} \quad \Delta p^{\text{CFD}}$$

In vitro



$$\Delta p^{\text{PINN}} \quad \Delta p^{\text{vWERP}} \quad \Delta p^{\text{Cath.}}$$

In vivo



$$\Delta p^{\text{PINN}} \quad \Delta p^{\text{vWERP}} \quad \Delta p^{\text{Cath.}}$$